How does a geosynchronous earth-imaging satellite improve continuous environmental and disaster monitoring over India?

GS Paper III
Science and Tech
Disaster Management

KNOWLEDGE POINTS

- **Fixed vantage point:** By matching Earth's rotation at 36,000 kilometres altitude, the satellite stays focused over one region. This eliminates gaps between passes, letting scientists observe the same landscape without waiting days for the spacecraft to circle back.
- **Real-time disaster tracking:** The spacecraft watches fast-evolving emergencies like flash floods, cyclones, and forest fires as they unfold. Continuous imaging helps emergency teams detect sudden environmental shifts instantly and respond before natural crises cause widespread community damage.
- **Advanced spectral scanning:** Instead of simple photography, specialized sensors record reflected light across multiple wavelengths. This technical capability lets researchers clearly distinguish farm crops, water bodies, and vegetation health, exposing subtle ecological changes across seasons.
- **Orbit trade-offs:** Low-earth satellites circle a few hundred kilometres up, capturing very sharp images but taking days to revisit any area. In contrast, higher geosynchronous satellites sacrifice pinpoint detail to deliver continuous regional views.
- **Preventing monitoring gaps:** Farmers sometimes burn crop stubble at odd hours to evade low-orbit satellites that pass only at fixed times. Constant observation ensures continuous monitoring, making it impossible for polluters to hide smoke emissions.

QUICK FACTS

- The EOS-05 spacecraft will orbit Earth at approximately 36,000 kilometres in geosynchronous orbit, matching the planet's rotation to provide uninterrupted observations over the Indian subcontinent.
- Recent flash floods from the Bhote Koshi deluge in Nepal caused over 1,344 deaths, highlighting the urgent need for persistent space-based monitoring of fragile Himalayan landscapes.
- Unlike dedicated land-imaging satellites that historically operated only in low orbits, India previously maintained only weather monitoring satellites within geostationary orbit under the INSAT-3D series.
- Conventional low-earth imaging satellites fly just a few hundred kilometres high, capturing razor-sharp details but requiring several days to complete successive orbital passes over identical terrain.

KEY TAKEAWAY

By operating continuously from geosynchronous orbit, EOS-05 replaces intermittent satellite passes with uninterrupted spectral monitoring, giving disaster responders and environmental managers instant views of rapid ground changes across India.

How has the Strait of Hormuz trapped the United States in an unwinnable regional conflict?

GS Paper II
Effect of Policies Developed and Developing Countries

KNOWLEDGE POINTS

- **Strategic battlefield shift:** Tehran turned the narrow waterway into the main arena of war. This move completely shifted focus away from America's original goals of destroying nuclear sites or changing Iran's government, forcing Washington into a frustrating tactical maritime battle.
- **Asymmetric deterrence power:** Iran repeatedly targets commercial ships and military vessels using geography to its advantage. Even heavy economic sanctions and naval blockades have failed to break Iran's chokehold, leaving America unable to secure the shipping lane or claim decisive victory.
- **Regional base vulnerability:** Whenever American forces strike Iranian targets or tankers, Iran retaliates directly against American military bases in Kuwait, Bahrain, and Jordan. This pushback raises the human and material cost of the war, trapping American forces in endless counter-strikes.
- **Mowing the lawn:** Instead of achieving a decisive political breakthrough, Washington engages in repetitive strikes simply to blunt Iranian capabilities. This attritional cycle never eliminates underlying threats, as each wave of attacks provokes bolder Iranian responses.
- **Failed peace diplomacy:** The collapse of the June 17 Islamabad agreement removed formal diplomatic guardrails. Without open communication channels or agreed ceasefire terms, minor tactical clashes at sea quickly escalate into wider regional conflict.

QUICK FACTS

- The current conflict began on February 28, initially launched by the United States and Israel with the declared goals of neutralizing Iran’s nuclear capabilities and seeking regime change.
- Following the breakdown of the June 17 Islamabad Memorandum of Understanding, bilateral diplomacy collapsed, returning both nations to direct naval warfare and port blockades across Gulf waterways.
- Retaliating against American strikes on Iranian oil tankers, the Islamic Revolutionary Guard Corps launched strikes against six maritime vessels, directly targeting three commercial tankers and three U.S.-linked ships.
- The Strait of Hormuz forms the primary maritime chokepoint linking the Persian Gulf with the Sea of Oman, normally handling approximately one-fifth of total global petroleum consumption.

KEY TAKEAWAY

By turning the Strait of Hormuz into an asymmetric battleground, Iran deflected Washington's primary war objectives and trapped American forces in an escalating, open-ended attritional conflict without any clear exit strategy.